AMReX/incflo

Open Source CFD Solver

What is incflo?

- Solves for the velocity and pressure of an incompressible fluid
- 3D Objects can be inserted to simulate how this affects flow
- Numerically solves the Navier-Stokes equations
- Developed primarily at Lawrence Berkeley National Laboratory
- CFD is used primarily by aerospace and mechanical engineers

What makes incflo different?

- Focuses on variable density fluids
  - Focus on fluid mixing
- Uses cut-cells to model 3D objects
  - Computationally less expensive than importing a mesh
- Uses Adaptive Mesh Refinement (AMR)
  - Adds more cells where there’s complex geometry

How does the algorithm work?

- Calculations are performed iteratively over multiple time steps
- For each time step:
  - **Step 1:** Calculator velocity using Navier-Stokes without pressure term
  - **Step 2:** Use a linear system to calculate the pressure in each cell
  - **Step 3:** Correct velocity using pressure term

Step 1: Provisional Velocity

Full Momentum Equation

$$\frac{\partial \mathbf{U}}{\partial t} = -(\mathbf{U} \cdot \nabla)\mathbf{U} - \frac{1}{\rho}\nabla p + \frac{1}{\rho}\nabla \cdot \boldsymbol{\tau} + \frac{\mathbf{f}}{\rho}$$

Equation of each time step

$$\frac{\mathbf{U}^* - \mathbf{U}^n}{\Delta t} = -(\mathbf{U}^n \cdot \nabla)\mathbf{U}^n + \frac{1}{\rho^n}\nabla \cdot \boldsymbol{\tau}^n + \mathbf{g} + \frac{\mathbf{F}^n}{\rho^n}$$

- Gradients computed by performing calculations with velocity of neighbors
- Program runs calculations in parallel to optimize performance

Step 2: Calculate Pressure

Using the below equation,we can solve for the pressure (ϕ) at each cell.

$$\nabla \cdot \left(\frac{1}{\rho(\mathbf{x})} \nabla \phi(\mathbf{x}) \right) = \nabla \cdot \mathbf{U}^*(\mathbf{x})$$

- We can turn the above equation in the form Ax = b to solve as a linear equation.
- Linear system is adjusted to account for the size of cell and amount of fluid in cell
- Uses Multi-Level Multi-Grid (MLMG) solver to dampen errors across course and fine grid hierarchies to achieve O(n)
- Uses local cell data instead of matrices to perform calculations

Step 3: Correct Velocity

Use the calculated pressure to correct the provisional velocity

$$\mathbf{U}^{n+1} = \mathbf{U}^{*} - \frac{\Delta t}{\rho}\nabla\phi$$

Accuracy and Conditioning

- For some test simulations, numerical outputs are compared to known analytical solutions at each time step
- This solver is mean to be more accurate since it accounts for variable density
- Extremely small cut cells or complex geometries can lead to ill-conditioning
- The software will redistribute how smaller cells are calculated to reduce ill-conditioning
- MLMG helps to avoid ill conditioning of extremely large matrices

Question and Experiment

- Question: What algorithms and computational expenses are involved in creating the cut cells?
- Experiment: How does the speed of a fluid vary over time for a particular point depending on the inserted geometry?